SOLVED PRACTICE WORKBOOK

Islands and Coral Reefs / India Islands-Great Nicobar - Solved Practice Workbook

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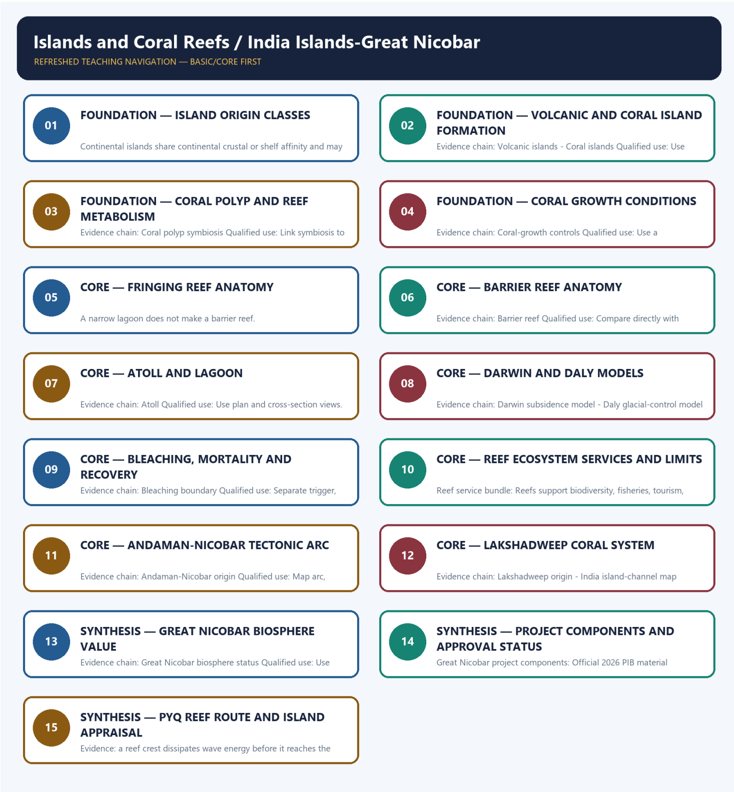

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BASIC MCQS / REMEDIATION

Q1. Which statement correctly explains Island classification?

A. Islands may be continental fragments or shelf islands, oceanic volcanic islands, coral islands or depositional islands; classification must state the origin criterion. B. Coral islands are low accumulations of reef-derived carbonate sediment built and reworked by organisms, waves and currents; the living reef and the island landform are related but distinct. C. Oceanic volcanic islands grow where magma builds edifices above sea level at plate boundaries or hotspots; volcanic origin does not imply current eruptive activity. D. Continental islands share continental crustal or shelf affinity and may be separated by drowning, faulting or erosion; proximity to a continent alone does not prove origin.

Answer: A. Explanation: Islands may be continental fragments or shelf islands, oceanic volcanic islands, coral islands or depositional islands; classification must state the origin criterion. The other options describe different processes, locations, scales or governance categories.

Q2. Which option is the safest spatial interpretation of Island classification?

A. Oceanic volcanic islands grow where magma builds edifices above sea level at plate boundaries or hotspots; volcanic origin does not imply current eruptive activity. B. Islands may be continental fragments or shelf islands, oceanic volcanic islands, coral islands or depositional islands; classification must state the origin criterion. C. Reef-building corals are animals that secrete calcium-carbonate skeletons and commonly host photosynthetic zooxanthellae, linking clear sunlit water to reef productivity. D. Coral islands are low accumulations of reef-derived carbonate sediment built and reworked by organisms, waves and currents; the living reef and the island landform are related but distinct.

Answer: B. Explanation: Islands may be continental fragments or shelf islands, oceanic volcanic islands, coral islands or depositional islands; classification must state the origin criterion. The other options describe different processes, locations, scales or governance categories.

Q3. Which statement preserves the process boundary for Island classification?

A. Coral islands are low accumulations of reef-derived carbonate sediment built and reworked by organisms, waves and currents; the living reef and the island landform are related but distinct. B. Most reef-building corals favour warm, shallow, clear, well-lit, normally saline and oxygenated water with limited sediment stress; local tolerances vary. C. Islands may be continental fragments or shelf islands, oceanic volcanic islands, coral islands or depositional islands; classification must state the origin criterion. D. Reef-building corals are animals that secrete calcium-carbonate skeletons and commonly host photosynthetic zooxanthellae, linking clear sunlit water to reef productivity.

Answer: C. Explanation: Islands may be continental fragments or shelf islands, oceanic volcanic islands, coral islands or depositional islands; classification must state the origin criterion. The other options describe different processes, locations, scales or governance categories.

Q4. Which option avoids the main UPSC trap concerning Island classification?

A. A fringing reef lies close to shore with no lagoon or only a narrow shallow lagoon, making it the closest reef type to the land. B. Most reef-building corals favour warm, shallow, clear, well-lit, normally saline and oxygenated water with limited sediment stress; local tolerances vary. C. Reef-building corals are animals that secrete calcium-carbonate skeletons and commonly host photosynthetic zooxanthellae, linking clear sunlit water to reef productivity. D. Islands may be continental fragments or shelf islands, oceanic volcanic islands, coral islands or depositional islands; classification must state the origin criterion.

Answer: D. Explanation: Islands may be continental fragments or shelf islands, oceanic volcanic islands, coral islands or depositional islands; classification must state the origin criterion. The other options describe different processes, locations, scales or governance categories.

Q5. Which statement correctly explains Continental islands?

A. Continental islands share continental crustal or shelf affinity and may be separated by drowning, faulting or erosion; proximity to a continent alone does not prove origin. B. Coral islands are low accumulations of reef-derived carbonate sediment built and reworked by organisms, waves and currents; the living reef and the island landform are related but distinct. C. Reef-building corals are animals that secrete calcium-carbonate skeletons and commonly host photosynthetic zooxanthellae, linking clear sunlit water to reef productivity. D. Oceanic volcanic islands grow where magma builds edifices above sea level at plate boundaries or hotspots; volcanic origin does not imply current eruptive activity.

Answer: A. Explanation: Continental islands share continental crustal or shelf affinity and may be separated by drowning, faulting or erosion; proximity to a continent alone does not prove origin. The other options describe different processes, locations, scales or governance categories.

Q6. Which option is the safest spatial interpretation of Continental islands?

A. Most reef-building corals favour warm, shallow, clear, well-lit, normally saline and oxygenated water with limited sediment stress; local tolerances vary. B. Continental islands share continental crustal or shelf affinity and may be separated by drowning, faulting or erosion; proximity to a continent alone does not prove origin. C. Coral islands are low accumulations of reef-derived carbonate sediment built and reworked by organisms, waves and currents; the living reef and the island landform are related but distinct. D. Reef-building corals are animals that secrete calcium-carbonate skeletons and commonly host photosynthetic zooxanthellae, linking clear sunlit water to reef productivity.

Answer: B. Explanation: Continental islands share continental crustal or shelf affinity and may be separated by drowning, faulting or erosion; proximity to a continent alone does not prove origin. The other options describe different processes, locations, scales or governance categories.

Q7. Which statement preserves the process boundary for Continental islands?

A. A fringing reef lies close to shore with no lagoon or only a narrow shallow lagoon, making it the closest reef type to the land. B. Reef-building corals are animals that secrete calcium-carbonate skeletons and commonly host photosynthetic zooxanthellae, linking clear sunlit water to reef productivity. C. Continental islands share continental crustal or shelf affinity and may be separated by drowning, faulting or erosion; proximity to a continent alone does not prove origin. D. Most reef-building corals favour warm, shallow, clear, well-lit, normally saline and oxygenated water with limited sediment stress; local tolerances vary.

Answer: C. Explanation: Continental islands share continental crustal or shelf affinity and may be separated by drowning, faulting or erosion; proximity to a continent alone does not prove origin. The other options describe different processes, locations, scales or governance categories.

Q8. Which option avoids the main UPSC trap concerning Continental islands?

A. A fringing reef lies close to shore with no lagoon or only a narrow shallow lagoon, making it the closest reef type to the land. B. Most reef-building corals favour warm, shallow, clear, well-lit, normally saline and oxygenated water with limited sediment stress; local tolerances vary. C. A barrier reef lies farther offshore and is separated from land by a broader, deeper lagoon; offshore position and lagoon width distinguish it from a fringing reef. D. Continental islands share continental crustal or shelf affinity and may be separated by drowning, faulting or erosion; proximity to a continent alone does not prove origin.

Answer: D. Explanation: Continental islands share continental crustal or shelf affinity and may be separated by drowning, faulting or erosion; proximity to a continent alone does not prove origin. The other options describe different processes, locations, scales or governance categories.

Q9. Which statement correctly explains Volcanic islands?

A. Oceanic volcanic islands grow where magma builds edifices above sea level at plate boundaries or hotspots; volcanic origin does not imply current eruptive activity. B. Most reef-building corals favour warm, shallow, clear, well-lit, normally saline and oxygenated water with limited sediment stress; local tolerances vary. C. Coral islands are low accumulations of reef-derived carbonate sediment built and reworked by organisms, waves and currents; the living reef and the island landform are related but distinct. D. Reef-building corals are animals that secrete calcium-carbonate skeletons and commonly host photosynthetic zooxanthellae, linking clear sunlit water to reef productivity.

Answer: A. Explanation: Oceanic volcanic islands grow where magma builds edifices above sea level at plate boundaries or hotspots; volcanic origin does not imply current eruptive activity. The other options describe different processes, locations, scales or governance categories.

Q10. Which option is the safest spatial interpretation of Volcanic islands?

A. Most reef-building corals favour warm, shallow, clear, well-lit, normally saline and oxygenated water with limited sediment stress; local tolerances vary. B. Oceanic volcanic islands grow where magma builds edifices above sea level at plate boundaries or hotspots; volcanic origin does not imply current eruptive activity. C. A fringing reef lies close to shore with no lagoon or only a narrow shallow lagoon, making it the closest reef type to the land. D. Reef-building corals are animals that secrete calcium-carbonate skeletons and commonly host photosynthetic zooxanthellae, linking clear sunlit water to reef productivity.

Answer: B. Explanation: Oceanic volcanic islands grow where magma builds edifices above sea level at plate boundaries or hotspots; volcanic origin does not imply current eruptive activity. The other options describe different processes, locations, scales or governance categories.

Q11. Which statement preserves the process boundary for Volcanic islands?

A. A barrier reef lies farther offshore and is separated from land by a broader, deeper lagoon; offshore position and lagoon width distinguish it from a fringing reef. B. A fringing reef lies close to shore with no lagoon or only a narrow shallow lagoon, making it the closest reef type to the land. C. Oceanic volcanic islands grow where magma builds edifices above sea level at plate boundaries or hotspots; volcanic origin does not imply current eruptive activity. D. Most reef-building corals favour warm, shallow, clear, well-lit, normally saline and oxygenated water with limited sediment stress; local tolerances vary.

Answer: C. Explanation: Oceanic volcanic islands grow where magma builds edifices above sea level at plate boundaries or hotspots; volcanic origin does not imply current eruptive activity. The other options describe different processes, locations, scales or governance categories.

Q12. Which option avoids the main UPSC trap concerning Volcanic islands?

A. An atoll is a roughly ring-shaped reef or reef-island system enclosing a lagoon, generally without a central high island at the surface. B. A fringing reef lies close to shore with no lagoon or only a narrow shallow lagoon, making it the closest reef type to the land. C. A barrier reef lies farther offshore and is separated from land by a broader, deeper lagoon; offshore position and lagoon width distinguish it from a fringing reef. D. Oceanic volcanic islands grow where magma builds edifices above sea level at plate boundaries or hotspots; volcanic origin does not imply current eruptive activity.

Answer: D. Explanation: Oceanic volcanic islands grow where magma builds edifices above sea level at plate boundaries or hotspots; volcanic origin does not imply current eruptive activity. The other options describe different processes, locations, scales or governance categories.

Q13. Which statement correctly explains Coral islands?

A. Coral islands are low accumulations of reef-derived carbonate sediment built and reworked by organisms, waves and currents; the living reef and the island landform are related but distinct. B. Reef-building corals are animals that secrete calcium-carbonate skeletons and commonly host photosynthetic zooxanthellae, linking clear sunlit water to reef productivity. C. Most reef-building corals favour warm, shallow, clear, well-lit, normally saline and oxygenated water with limited sediment stress; local tolerances vary. D. A fringing reef lies close to shore with no lagoon or only a narrow shallow lagoon, making it the closest reef type to the land.

Answer: A. Explanation: Coral islands are low accumulations of reef-derived carbonate sediment built and reworked by organisms, waves and currents; the living reef and the island landform are related but distinct. The other options describe different processes, locations, scales or governance categories.

Q14. Which option is the safest spatial interpretation of Coral islands?

A. A fringing reef lies close to shore with no lagoon or only a narrow shallow lagoon, making it the closest reef type to the land. B. Coral islands are low accumulations of reef-derived carbonate sediment built and reworked by organisms, waves and currents; the living reef and the island landform are related but distinct. C. A barrier reef lies farther offshore and is separated from land by a broader, deeper lagoon; offshore position and lagoon width distinguish it from a fringing reef. D. Most reef-building corals favour warm, shallow, clear, well-lit, normally saline and oxygenated water with limited sediment stress; local tolerances vary.

Answer: B. Explanation: Coral islands are low accumulations of reef-derived carbonate sediment built and reworked by organisms, waves and currents; the living reef and the island landform are related but distinct. The other options describe different processes, locations, scales or governance categories.

Q15. Which statement preserves the process boundary for Coral islands?

A. A barrier reef lies farther offshore and is separated from land by a broader, deeper lagoon; offshore position and lagoon width distinguish it from a fringing reef. B. A fringing reef lies close to shore with no lagoon or only a narrow shallow lagoon, making it the closest reef type to the land. C. Coral islands are low accumulations of reef-derived carbonate sediment built and reworked by organisms, waves and currents; the living reef and the island landform are related but distinct. D. An atoll is a roughly ring-shaped reef or reef-island system enclosing a lagoon, generally without a central high island at the surface.

Answer: C. Explanation: Coral islands are low accumulations of reef-derived carbonate sediment built and reworked by organisms, waves and currents; the living reef and the island landform are related but distinct. The other options describe different processes, locations, scales or governance categories.

Q16. Which option avoids the main UPSC trap concerning Coral islands?

A. Darwin's model links fringing reef to barrier reef to atoll as a volcanic foundation slowly subsides while coral growth keeps pace near sea level; it is a process model, not a rule for every reef. B. A barrier reef lies farther offshore and is separated from land by a broader, deeper lagoon; offshore position and lagoon width distinguish it from a fringing reef. C. An atoll is a roughly ring-shaped reef or reef-island system enclosing a lagoon, generally without a central high island at the surface. D. Coral islands are low accumulations of reef-derived carbonate sediment built and reworked by organisms, waves and currents; the living reef and the island landform are related but distinct.

Answer: D. Explanation: Coral islands are low accumulations of reef-derived carbonate sediment built and reworked by organisms, waves and currents; the living reef and the island landform are related but distinct. The other options describe different processes, locations, scales or governance categories.

Q17. Which statement correctly explains Coral polyp symbiosis?

A. Reef-building corals are animals that secrete calcium-carbonate skeletons and commonly host photosynthetic zooxanthellae, linking clear sunlit water to reef productivity. B. Most reef-building corals favour warm, shallow, clear, well-lit, normally saline and oxygenated water with limited sediment stress; local tolerances vary. C. A fringing reef lies close to shore with no lagoon or only a narrow shallow lagoon, making it the closest reef type to the land. D. A barrier reef lies farther offshore and is separated from land by a broader, deeper lagoon; offshore position and lagoon width distinguish it from a fringing reef.

Answer: A. Explanation: Reef-building corals are animals that secrete calcium-carbonate skeletons and commonly host photosynthetic zooxanthellae, linking clear sunlit water to reef productivity. The other options describe different processes, locations, scales or governance categories.

Q18. Which option is the safest spatial interpretation of Coral polyp symbiosis?

A. An atoll is a roughly ring-shaped reef or reef-island system enclosing a lagoon, generally without a central high island at the surface. B. Reef-building corals are animals that secrete calcium-carbonate skeletons and commonly host photosynthetic zooxanthellae, linking clear sunlit water to reef productivity. C. A barrier reef lies farther offshore and is separated from land by a broader, deeper lagoon; offshore position and lagoon width distinguish it from a fringing reef. D. A fringing reef lies close to shore with no lagoon or only a narrow shallow lagoon, making it the closest reef type to the land.

Answer: B. Explanation: Reef-building corals are animals that secrete calcium-carbonate skeletons and commonly host photosynthetic zooxanthellae, linking clear sunlit water to reef productivity. The other options describe different processes, locations, scales or governance categories.

Q19. Which statement preserves the process boundary for Coral polyp symbiosis?

A. An atoll is a roughly ring-shaped reef or reef-island system enclosing a lagoon, generally without a central high island at the surface. B. Darwin's model links fringing reef to barrier reef to atoll as a volcanic foundation slowly subsides while coral growth keeps pace near sea level; it is a process model, not a rule for every reef. C. Reef-building corals are animals that secrete calcium-carbonate skeletons and commonly host photosynthetic zooxanthellae, linking clear sunlit water to reef productivity. D. A barrier reef lies farther offshore and is separated from land by a broader, deeper lagoon; offshore position and lagoon width distinguish it from a fringing reef.

Answer: C. Explanation: Reef-building corals are animals that secrete calcium-carbonate skeletons and commonly host photosynthetic zooxanthellae, linking clear sunlit water to reef productivity. The other options describe different processes, locations, scales or governance categories.

Q20. Which option avoids the main UPSC trap concerning Coral polyp symbiosis?

A. Daly emphasised lowered glacial sea level, wave planation and post-glacial flooding as controls on reef platforms; it supplements rather than erases biological growth and tectonics. B. Darwin's model links fringing reef to barrier reef to atoll as a volcanic foundation slowly subsides while coral growth keeps pace near sea level; it is a process model, not a rule for every reef. C. An atoll is a roughly ring-shaped reef or reef-island system enclosing a lagoon, generally without a central high island at the surface. D. Reef-building corals are animals that secrete calcium-carbonate skeletons and commonly host photosynthetic zooxanthellae, linking clear sunlit water to reef productivity.

Answer: D. Explanation: Reef-building corals are animals that secrete calcium-carbonate skeletons and commonly host photosynthetic zooxanthellae, linking clear sunlit water to reef productivity. The other options describe different processes, locations, scales or governance categories.

Q21. Which statement correctly explains Coral-growth controls?

A. Most reef-building corals favour warm, shallow, clear, well-lit, normally saline and oxygenated water with limited sediment stress; local tolerances vary. B. An atoll is a roughly ring-shaped reef or reef-island system enclosing a lagoon, generally without a central high island at the surface. C. A fringing reef lies close to shore with no lagoon or only a narrow shallow lagoon, making it the closest reef type to the land. D. A barrier reef lies farther offshore and is separated from land by a broader, deeper lagoon; offshore position and lagoon width distinguish it from a fringing reef.

Answer: A. Explanation: Most reef-building corals favour warm, shallow, clear, well-lit, normally saline and oxygenated water with limited sediment stress; local tolerances vary. The other options describe different processes, locations, scales or governance categories.

Q22. Which option is the safest spatial interpretation of Coral-growth controls?

A. Darwin's model links fringing reef to barrier reef to atoll as a volcanic foundation slowly subsides while coral growth keeps pace near sea level; it is a process model, not a rule for every reef. B. Most reef-building corals favour warm, shallow, clear, well-lit, normally saline and oxygenated water with limited sediment stress; local tolerances vary. C. A barrier reef lies farther offshore and is separated from land by a broader, deeper lagoon; offshore position and lagoon width distinguish it from a fringing reef. D. An atoll is a roughly ring-shaped reef or reef-island system enclosing a lagoon, generally without a central high island at the surface.

Answer: B. Explanation: Most reef-building corals favour warm, shallow, clear, well-lit, normally saline and oxygenated water with limited sediment stress; local tolerances vary. The other options describe different processes, locations, scales or governance categories.

Q23. Which statement preserves the process boundary for Coral-growth controls?

A. Daly emphasised lowered glacial sea level, wave planation and post-glacial flooding as controls on reef platforms; it supplements rather than erases biological growth and tectonics. B. Darwin's model links fringing reef to barrier reef to atoll as a volcanic foundation slowly subsides while coral growth keeps pace near sea level; it is a process model, not a rule for every reef. C. Most reef-building corals favour warm, shallow, clear, well-lit, normally saline and oxygenated water with limited sediment stress; local tolerances vary. D. An atoll is a roughly ring-shaped reef or reef-island system enclosing a lagoon, generally without a central high island at the surface.

Answer: C. Explanation: Most reef-building corals favour warm, shallow, clear, well-lit, normally saline and oxygenated water with limited sediment stress; local tolerances vary. The other options describe different processes, locations, scales or governance categories.

Q24. Which option avoids the main UPSC trap concerning Coral-growth controls?

A. Coral bleaching is the loss of symbiotic algae or pigments under stress, often marine heat; bleaching raises mortality risk but does not mean every colony is already dead. B. Daly emphasised lowered glacial sea level, wave planation and post-glacial flooding as controls on reef platforms; it supplements rather than erases biological growth and tectonics. C. Darwin's model links fringing reef to barrier reef to atoll as a volcanic foundation slowly subsides while coral growth keeps pace near sea level; it is a process model, not a rule for every reef. D. Most reef-building corals favour warm, shallow, clear, well-lit, normally saline and oxygenated water with limited sediment stress; local tolerances vary.

Answer: D. Explanation: Most reef-building corals favour warm, shallow, clear, well-lit, normally saline and oxygenated water with limited sediment stress; local tolerances vary. The other options describe different processes, locations, scales or governance categories.

Q25. Which statement correctly explains Fringing reef?

A. A fringing reef lies close to shore with no lagoon or only a narrow shallow lagoon, making it the closest reef type to the land. B. An atoll is a roughly ring-shaped reef or reef-island system enclosing a lagoon, generally without a central high island at the surface. C. A barrier reef lies farther offshore and is separated from land by a broader, deeper lagoon; offshore position and lagoon width distinguish it from a fringing reef. D. Darwin's model links fringing reef to barrier reef to atoll as a volcanic foundation slowly subsides while coral growth keeps pace near sea level; it is a process model, not a rule for every reef.

Answer: A. Explanation: A fringing reef lies close to shore with no lagoon or only a narrow shallow lagoon, making it the closest reef type to the land. The other options describe different processes, locations, scales or governance categories.

Q26. Which option is the safest spatial interpretation of Fringing reef?

A. Darwin's model links fringing reef to barrier reef to atoll as a volcanic foundation slowly subsides while coral growth keeps pace near sea level; it is a process model, not a rule for every reef. B. A fringing reef lies close to shore with no lagoon or only a narrow shallow lagoon, making it the closest reef type to the land. C. Daly emphasised lowered glacial sea level, wave planation and post-glacial flooding as controls on reef platforms; it supplements rather than erases biological growth and tectonics. D. An atoll is a roughly ring-shaped reef or reef-island system enclosing a lagoon, generally without a central high island at the surface.

Answer: B. Explanation: A fringing reef lies close to shore with no lagoon or only a narrow shallow lagoon, making it the closest reef type to the land. The other options describe different processes, locations, scales or governance categories.

Q27. Which statement preserves the process boundary for Fringing reef?

A. Darwin's model links fringing reef to barrier reef to atoll as a volcanic foundation slowly subsides while coral growth keeps pace near sea level; it is a process model, not a rule for every reef. B. Coral bleaching is the loss of symbiotic algae or pigments under stress, often marine heat; bleaching raises mortality risk but does not mean every colony is already dead. C. A fringing reef lies close to shore with no lagoon or only a narrow shallow lagoon, making it the closest reef type to the land. D. Daly emphasised lowered glacial sea level, wave planation and post-glacial flooding as controls on reef platforms; it supplements rather than erases biological growth and tectonics.

Answer: C. Explanation: A fringing reef lies close to shore with no lagoon or only a narrow shallow lagoon, making it the closest reef type to the land. The other options describe different processes, locations, scales or governance categories.

Q28. Which option avoids the main UPSC trap concerning Fringing reef?

A. Reefs support biodiversity, fisheries, tourism, carbonate sediment and wave-energy attenuation, but they cannot eliminate extreme surge, tsunami or poor land-use exposure. B. Coral bleaching is the loss of symbiotic algae or pigments under stress, often marine heat; bleaching raises mortality risk but does not mean every colony is already dead. C. Daly emphasised lowered glacial sea level, wave planation and post-glacial flooding as controls on reef platforms; it supplements rather than erases biological growth and tectonics. D. A fringing reef lies close to shore with no lagoon or only a narrow shallow lagoon, making it the closest reef type to the land.

Answer: D. Explanation: A fringing reef lies close to shore with no lagoon or only a narrow shallow lagoon, making it the closest reef type to the land. The other options describe different processes, locations, scales or governance categories.

Q29. Which statement correctly explains Barrier reef?

A. A barrier reef lies farther offshore and is separated from land by a broader, deeper lagoon; offshore position and lagoon width distinguish it from a fringing reef. B. An atoll is a roughly ring-shaped reef or reef-island system enclosing a lagoon, generally without a central high island at the surface. C. Darwin's model links fringing reef to barrier reef to atoll as a volcanic foundation slowly subsides while coral growth keeps pace near sea level; it is a process model, not a rule for every reef. D. Daly emphasised lowered glacial sea level, wave planation and post-glacial flooding as controls on reef platforms; it supplements rather than erases biological growth and tectonics.

Answer: A. Explanation: A barrier reef lies farther offshore and is separated from land by a broader, deeper lagoon; offshore position and lagoon width distinguish it from a fringing reef. The other options describe different processes, locations, scales or governance categories.

Q30. Which option is the safest spatial interpretation of Barrier reef?

A. Coral bleaching is the loss of symbiotic algae or pigments under stress, often marine heat; bleaching raises mortality risk but does not mean every colony is already dead. B. A barrier reef lies farther offshore and is separated from land by a broader, deeper lagoon; offshore position and lagoon width distinguish it from a fringing reef. C. Darwin's model links fringing reef to barrier reef to atoll as a volcanic foundation slowly subsides while coral growth keeps pace near sea level; it is a process model, not a rule for every reef. D. Daly emphasised lowered glacial sea level, wave planation and post-glacial flooding as controls on reef platforms; it supplements rather than erases biological growth and tectonics.

Answer: B. Explanation: A barrier reef lies farther offshore and is separated from land by a broader, deeper lagoon; offshore position and lagoon width distinguish it from a fringing reef. The other options describe different processes, locations, scales or governance categories.

Q31. Which statement preserves the process boundary for Barrier reef?

A. Reefs support biodiversity, fisheries, tourism, carbonate sediment and wave-energy attenuation, but they cannot eliminate extreme surge, tsunami or poor land-use exposure. B. Coral bleaching is the loss of symbiotic algae or pigments under stress, often marine heat; bleaching raises mortality risk but does not mean every colony is already dead. C. A barrier reef lies farther offshore and is separated from land by a broader, deeper lagoon; offshore position and lagoon width distinguish it from a fringing reef. D. Daly emphasised lowered glacial sea level, wave planation and post-glacial flooding as controls on reef platforms; it supplements rather than erases biological growth and tectonics.

Answer: C. Explanation: A barrier reef lies farther offshore and is separated from land by a broader, deeper lagoon; offshore position and lagoon width distinguish it from a fringing reef. The other options describe different processes, locations, scales or governance categories.

Q32. Which option avoids the main UPSC trap concerning Barrier reef?

A. Reefs support biodiversity, fisheries, tourism, carbonate sediment and wave-energy attenuation, but they cannot eliminate extreme surge, tsunami or poor land-use exposure. B. The Andaman and Nicobar Islands are emergent parts of a tectonic island arc connected to the Sunda-Andaman subduction system; the group is not simply a chain of coral atolls. C. Coral bleaching is the loss of symbiotic algae or pigments under stress, often marine heat; bleaching raises mortality risk but does not mean every colony is already dead. D. A barrier reef lies farther offshore and is separated from land by a broader, deeper lagoon; offshore position and lagoon width distinguish it from a fringing reef.

Answer: D. Explanation: A barrier reef lies farther offshore and is separated from land by a broader, deeper lagoon; offshore position and lagoon width distinguish it from a fringing reef. The other options describe different processes, locations, scales or governance categories.

Q33. Which statement correctly explains Atoll?

A. An atoll is a roughly ring-shaped reef or reef-island system enclosing a lagoon, generally without a central high island at the surface. B. Darwin's model links fringing reef to barrier reef to atoll as a volcanic foundation slowly subsides while coral growth keeps pace near sea level; it is a process model, not a rule for every reef. C. Coral bleaching is the loss of symbiotic algae or pigments under stress, often marine heat; bleaching raises mortality risk but does not mean every colony is already dead. D. Daly emphasised lowered glacial sea level, wave planation and post-glacial flooding as controls on reef platforms; it supplements rather than erases biological growth and tectonics.

Answer: A. Explanation: An atoll is a roughly ring-shaped reef or reef-island system enclosing a lagoon, generally without a central high island at the surface. The other options describe different processes, locations, scales or governance categories.

Q34. Which option is the safest spatial interpretation of Atoll?

A. Daly emphasised lowered glacial sea level, wave planation and post-glacial flooding as controls on reef platforms; it supplements rather than erases biological growth and tectonics. B. An atoll is a roughly ring-shaped reef or reef-island system enclosing a lagoon, generally without a central high island at the surface. C. Coral bleaching is the loss of symbiotic algae or pigments under stress, often marine heat; bleaching raises mortality risk but does not mean every colony is already dead. D. Reefs support biodiversity, fisheries, tourism, carbonate sediment and wave-energy attenuation, but they cannot eliminate extreme surge, tsunami or poor land-use exposure.

Answer: B. Explanation: An atoll is a roughly ring-shaped reef or reef-island system enclosing a lagoon, generally without a central high island at the surface. The other options describe different processes, locations, scales or governance categories.

Q35. Which statement preserves the process boundary for Atoll?

A. Coral bleaching is the loss of symbiotic algae or pigments under stress, often marine heat; bleaching raises mortality risk but does not mean every colony is already dead. B. The Andaman and Nicobar Islands are emergent parts of a tectonic island arc connected to the Sunda-Andaman subduction system; the group is not simply a chain of coral atolls. C. An atoll is a roughly ring-shaped reef or reef-island system enclosing a lagoon, generally without a central high island at the surface. D. Reefs support biodiversity, fisheries, tourism, carbonate sediment and wave-energy attenuation, but they cannot eliminate extreme surge, tsunami or poor land-use exposure.

Answer: C. Explanation: An atoll is a roughly ring-shaped reef or reef-island system enclosing a lagoon, generally without a central high island at the surface. The other options describe different processes, locations, scales or governance categories.

Q36. Which option avoids the main UPSC trap concerning Atoll?

A. Lakshadweep consists of low coral islands, reefs and atolls on submarine ridges in the Arabian Sea; reef-derived land and freshwater lenses make the islands physically sensitive. B. The Andaman and Nicobar Islands are emergent parts of a tectonic island arc connected to the Sunda-Andaman subduction system; the group is not simply a chain of coral atolls. C. Reefs support biodiversity, fisheries, tourism, carbonate sediment and wave-energy attenuation, but they cannot eliminate extreme surge, tsunami or poor land-use exposure. D. An atoll is a roughly ring-shaped reef or reef-island system enclosing a lagoon, generally without a central high island at the surface.

Answer: D. Explanation: An atoll is a roughly ring-shaped reef or reef-island system enclosing a lagoon, generally without a central high island at the surface. The other options describe different processes, locations, scales or governance categories.

Q37. Which statement correctly explains Darwin subsidence model?

A. Darwin's model links fringing reef to barrier reef to atoll as a volcanic foundation slowly subsides while coral growth keeps pace near sea level; it is a process model, not a rule for every reef. B. Daly emphasised lowered glacial sea level, wave planation and post-glacial flooding as controls on reef platforms; it supplements rather than erases biological growth and tectonics. C. Coral bleaching is the loss of symbiotic algae or pigments under stress, often marine heat; bleaching raises mortality risk but does not mean every colony is already dead. D. Reefs support biodiversity, fisheries, tourism, carbonate sediment and wave-energy attenuation, but they cannot eliminate extreme surge, tsunami or poor land-use exposure.

Answer: A. Explanation: Darwin's model links fringing reef to barrier reef to atoll as a volcanic foundation slowly subsides while coral growth keeps pace near sea level; it is a process model, not a rule for every reef. The other options describe different processes, locations, scales or governance categories.

Q38. Which option is the safest spatial interpretation of Darwin subsidence model?

A. Coral bleaching is the loss of symbiotic algae or pigments under stress, often marine heat; bleaching raises mortality risk but does not mean every colony is already dead. B. Darwin's model links fringing reef to barrier reef to atoll as a volcanic foundation slowly subsides while coral growth keeps pace near sea level; it is a process model, not a rule for every reef. C. The Andaman and Nicobar Islands are emergent parts of a tectonic island arc connected to the Sunda-Andaman subduction system; the group is not simply a chain of coral atolls. D. Reefs support biodiversity, fisheries, tourism, carbonate sediment and wave-energy attenuation, but they cannot eliminate extreme surge, tsunami or poor land-use exposure.

Answer: B. Explanation: Darwin's model links fringing reef to barrier reef to atoll as a volcanic foundation slowly subsides while coral growth keeps pace near sea level; it is a process model, not a rule for every reef. The other options describe different processes, locations, scales or governance categories.

Q39. Which statement preserves the process boundary for Darwin subsidence model?

A. Lakshadweep consists of low coral islands, reefs and atolls on submarine ridges in the Arabian Sea; reef-derived land and freshwater lenses make the islands physically sensitive. B. Reefs support biodiversity, fisheries, tourism, carbonate sediment and wave-energy attenuation, but they cannot eliminate extreme surge, tsunami or poor land-use exposure. C. Darwin's model links fringing reef to barrier reef to atoll as a volcanic foundation slowly subsides while coral growth keeps pace near sea level; it is a process model, not a rule for every reef. D. The Andaman and Nicobar Islands are emergent parts of a tectonic island arc connected to the Sunda-Andaman subduction system; the group is not simply a chain of coral atolls.

Answer: C. Explanation: Darwin's model links fringing reef to barrier reef to atoll as a volcanic foundation slowly subsides while coral growth keeps pace near sea level; it is a process model, not a rule for every reef. The other options describe different processes, locations, scales or governance categories.

Q40. Which option avoids the main UPSC trap concerning Darwin subsidence model?

A. Lakshadweep consists of low coral islands, reefs and atolls on submarine ridges in the Arabian Sea; reef-derived land and freshwater lenses make the islands physically sensitive. B. The Ten Degree Channel separates the Andaman group from the Nicobar group, while the Nine Degree Channel separates Minicoy from the main Lakshadweep group. C. The Andaman and Nicobar Islands are emergent parts of a tectonic island arc connected to the Sunda-Andaman subduction system; the group is not simply a chain of coral atolls. D. Darwin's model links fringing reef to barrier reef to atoll as a volcanic foundation slowly subsides while coral growth keeps pace near sea level; it is a process model, not a rule for every reef.

Answer: D. Explanation: Darwin's model links fringing reef to barrier reef to atoll as a volcanic foundation slowly subsides while coral growth keeps pace near sea level; it is a process model, not a rule for every reef. The other options describe different processes, locations, scales or governance categories.

Q41. Which statement correctly explains Daly glacial-control model?

A. Daly emphasised lowered glacial sea level, wave planation and post-glacial flooding as controls on reef platforms; it supplements rather than erases biological growth and tectonics. B. Coral bleaching is the loss of symbiotic algae or pigments under stress, often marine heat; bleaching raises mortality risk but does not mean every colony is already dead. C. Reefs support biodiversity, fisheries, tourism, carbonate sediment and wave-energy attenuation, but they cannot eliminate extreme surge, tsunami or poor land-use exposure. D. The Andaman and Nicobar Islands are emergent parts of a tectonic island arc connected to the Sunda-Andaman subduction system; the group is not simply a chain of coral atolls.

Answer: A. Explanation: Daly emphasised lowered glacial sea level, wave planation and post-glacial flooding as controls on reef platforms; it supplements rather than erases biological growth and tectonics. The other options describe different processes, locations, scales or governance categories.

Q42. Which option is the safest spatial interpretation of Daly glacial-control model?

A. Reefs support biodiversity, fisheries, tourism, carbonate sediment and wave-energy attenuation, but they cannot eliminate extreme surge, tsunami or poor land-use exposure. B. Daly emphasised lowered glacial sea level, wave planation and post-glacial flooding as controls on reef platforms; it supplements rather than erases biological growth and tectonics. C. Lakshadweep consists of low coral islands, reefs and atolls on submarine ridges in the Arabian Sea; reef-derived land and freshwater lenses make the islands physically sensitive. D. The Andaman and Nicobar Islands are emergent parts of a tectonic island arc connected to the Sunda-Andaman subduction system; the group is not simply a chain of coral atolls.

Answer: B. Explanation: Daly emphasised lowered glacial sea level, wave planation and post-glacial flooding as controls on reef platforms; it supplements rather than erases biological growth and tectonics. The other options describe different processes, locations, scales or governance categories.

Q43. Which statement preserves the process boundary for Daly glacial-control model?

A. The Ten Degree Channel separates the Andaman group from the Nicobar group, while the Nine Degree Channel separates Minicoy from the main Lakshadweep group. B. Lakshadweep consists of low coral islands, reefs and atolls on submarine ridges in the Arabian Sea; reef-derived land and freshwater lenses make the islands physically sensitive. C. Daly emphasised lowered glacial sea level, wave planation and post-glacial flooding as controls on reef platforms; it supplements rather than erases biological growth and tectonics. D. The Andaman and Nicobar Islands are emergent parts of a tectonic island arc connected to the Sunda-Andaman subduction system; the group is not simply a chain of coral atolls.

Answer: C. Explanation: Daly emphasised lowered glacial sea level, wave planation and post-glacial flooding as controls on reef platforms; it supplements rather than erases biological growth and tectonics. The other options describe different processes, locations, scales or governance categories.

Q44. Which option avoids the main UPSC trap concerning Daly glacial-control model?

A. Lakshadweep consists of low coral islands, reefs and atolls on submarine ridges in the Arabian Sea; reef-derived land and freshwater lenses make the islands physically sensitive. B. The Ten Degree Channel separates the Andaman group from the Nicobar group, while the Nine Degree Channel separates Minicoy from the main Lakshadweep group. C. Great Nicobar is in UNESCO's World Network of Biosphere Reserves, added in 2013; biosphere-reserve recognition does not make the entire island an inviolate no-use zone. D. Daly emphasised lowered glacial sea level, wave planation and post-glacial flooding as controls on reef platforms; it supplements rather than erases biological growth and tectonics.

Answer: D. Explanation: Daly emphasised lowered glacial sea level, wave planation and post-glacial flooding as controls on reef platforms; it supplements rather than erases biological growth and tectonics. The other options describe different processes, locations, scales or governance categories.

Q45. Which statement correctly explains Bleaching boundary?

A. Coral bleaching is the loss of symbiotic algae or pigments under stress, often marine heat; bleaching raises mortality risk but does not mean every colony is already dead. B. The Andaman and Nicobar Islands are emergent parts of a tectonic island arc connected to the Sunda-Andaman subduction system; the group is not simply a chain of coral atolls. C. Reefs support biodiversity, fisheries, tourism, carbonate sediment and wave-energy attenuation, but they cannot eliminate extreme surge, tsunami or poor land-use exposure. D. Lakshadweep consists of low coral islands, reefs and atolls on submarine ridges in the Arabian Sea; reef-derived land and freshwater lenses make the islands physically sensitive.

Answer: A. Explanation: Coral bleaching is the loss of symbiotic algae or pigments under stress, often marine heat; bleaching raises mortality risk but does not mean every colony is already dead. The other options describe different processes, locations, scales or governance categories.

Q46. Which option is the safest spatial interpretation of Bleaching boundary?

A. The Andaman and Nicobar Islands are emergent parts of a tectonic island arc connected to the Sunda-Andaman subduction system; the group is not simply a chain of coral atolls. B. Coral bleaching is the loss of symbiotic algae or pigments under stress, often marine heat; bleaching raises mortality risk but does not mean every colony is already dead. C. The Ten Degree Channel separates the Andaman group from the Nicobar group, while the Nine Degree Channel separates Minicoy from the main Lakshadweep group. D. Lakshadweep consists of low coral islands, reefs and atolls on submarine ridges in the Arabian Sea; reef-derived land and freshwater lenses make the islands physically sensitive.

Answer: B. Explanation: Coral bleaching is the loss of symbiotic algae or pigments under stress, often marine heat; bleaching raises mortality risk but does not mean every colony is already dead. The other options describe different processes, locations, scales or governance categories.

Q47. Which statement preserves the process boundary for Bleaching boundary?

A. The Ten Degree Channel separates the Andaman group from the Nicobar group, while the Nine Degree Channel separates Minicoy from the main Lakshadweep group. B. Lakshadweep consists of low coral islands, reefs and atolls on submarine ridges in the Arabian Sea; reef-derived land and freshwater lenses make the islands physically sensitive. C. Coral bleaching is the loss of symbiotic algae or pigments under stress, often marine heat; bleaching raises mortality risk but does not mean every colony is already dead. D. Great Nicobar is in UNESCO's World Network of Biosphere Reserves, added in 2013; biosphere-reserve recognition does not make the entire island an inviolate no-use zone.

Answer: C. Explanation: Coral bleaching is the loss of symbiotic algae or pigments under stress, often marine heat; bleaching raises mortality risk but does not mean every colony is already dead. The other options describe different processes, locations, scales or governance categories.

Q48. Which option avoids the main UPSC trap concerning Bleaching boundary?

A. The Ten Degree Channel separates the Andaman group from the Nicobar group, while the Nine Degree Channel separates Minicoy from the main Lakshadweep group. B. Official 2026 PIB material describes an international container transshipment port, airport, township and power infrastructure as the proposed integrated project components. C. Great Nicobar is in UNESCO's World Network of Biosphere Reserves, added in 2013; biosphere-reserve recognition does not make the entire island an inviolate no-use zone. D. Coral bleaching is the loss of symbiotic algae or pigments under stress, often marine heat; bleaching raises mortality risk but does not mean every colony is already dead.

Answer: D. Explanation: Coral bleaching is the loss of symbiotic algae or pigments under stress, often marine heat; bleaching raises mortality risk but does not mean every colony is already dead. The other options describe different processes, locations, scales or governance categories.

Q49. Which statement correctly explains Reef service bundle?

A. Reefs support biodiversity, fisheries, tourism, carbonate sediment and wave-energy attenuation, but they cannot eliminate extreme surge, tsunami or poor land-use exposure. B. Lakshadweep consists of low coral islands, reefs and atolls on submarine ridges in the Arabian Sea; reef-derived land and freshwater lenses make the islands physically sensitive. C. The Ten Degree Channel separates the Andaman group from the Nicobar group, while the Nine Degree Channel separates Minicoy from the main Lakshadweep group. D. The Andaman and Nicobar Islands are emergent parts of a tectonic island arc connected to the Sunda-Andaman subduction system; the group is not simply a chain of coral atolls.

Answer: A. Explanation: Reefs support biodiversity, fisheries, tourism, carbonate sediment and wave-energy attenuation, but they cannot eliminate extreme surge, tsunami or poor land-use exposure. The other options describe different processes, locations, scales or governance categories.

Q50. Which option is the safest spatial interpretation of Reef service bundle?

A. Lakshadweep consists of low coral islands, reefs and atolls on submarine ridges in the Arabian Sea; reef-derived land and freshwater lenses make the islands physically sensitive. B. Reefs support biodiversity, fisheries, tourism, carbonate sediment and wave-energy attenuation, but they cannot eliminate extreme surge, tsunami or poor land-use exposure. C. Great Nicobar is in UNESCO's World Network of Biosphere Reserves, added in 2013; biosphere-reserve recognition does not make the entire island an inviolate no-use zone. D. The Ten Degree Channel separates the Andaman group from the Nicobar group, while the Nine Degree Channel separates Minicoy from the main Lakshadweep group.

Answer: B. Explanation: Reefs support biodiversity, fisheries, tourism, carbonate sediment and wave-energy attenuation, but they cannot eliminate extreme surge, tsunami or poor land-use exposure. The other options describe different processes, locations, scales or governance categories.

Q51. Which statement preserves the process boundary for Reef service bundle?

A. Official 2026 PIB material describes an international container transshipment port, airport, township and power infrastructure as the proposed integrated project components. B. The Ten Degree Channel separates the Andaman group from the Nicobar group, while the Nine Degree Channel separates Minicoy from the main Lakshadweep group. C. Reefs support biodiversity, fisheries, tourism, carbonate sediment and wave-energy attenuation, but they cannot eliminate extreme surge, tsunami or poor land-use exposure. D. Great Nicobar is in UNESCO's World Network of Biosphere Reserves, added in 2013; biosphere-reserve recognition does not make the entire island an inviolate no-use zone.

Answer: C. Explanation: Reefs support biodiversity, fisheries, tourism, carbonate sediment and wave-energy attenuation, but they cannot eliminate extreme surge, tsunami or poor land-use exposure. The other options describe different processes, locations, scales or governance categories.

Q52. Which option avoids the main UPSC trap concerning Reef service bundle?

A. Official 2026 PIB material describes an international container transshipment port, airport, township and power infrastructure as the proposed integrated project components. B. A Rajya Sabha answer dated 12 February 2026 stated that Stage-II or final forest approval had not been granted and project activity depended on final approval; a factsheet is not proof of completed mitigation. C. Great Nicobar is in UNESCO's World Network of Biosphere Reserves, added in 2013; biosphere-reserve recognition does not make the entire island an inviolate no-use zone. D. Reefs support biodiversity, fisheries, tourism, carbonate sediment and wave-energy attenuation, but they cannot eliminate extreme surge, tsunami or poor land-use exposure.

Answer: D. Explanation: Reefs support biodiversity, fisheries, tourism, carbonate sediment and wave-energy attenuation, but they cannot eliminate extreme surge, tsunami or poor land-use exposure. The other options describe different processes, locations, scales or governance categories.

Q53. Which statement correctly explains Andaman-Nicobar origin?

A. The Andaman and Nicobar Islands are emergent parts of a tectonic island arc connected to the Sunda-Andaman subduction system; the group is not simply a chain of coral atolls. B. The Ten Degree Channel separates the Andaman group from the Nicobar group, while the Nine Degree Channel separates Minicoy from the main Lakshadweep group. C. Great Nicobar is in UNESCO's World Network of Biosphere Reserves, added in 2013; biosphere-reserve recognition does not make the entire island an inviolate no-use zone. D. Lakshadweep consists of low coral islands, reefs and atolls on submarine ridges in the Arabian Sea; reef-derived land and freshwater lenses make the islands physically sensitive.

Answer: A. Explanation: The Andaman and Nicobar Islands are emergent parts of a tectonic island arc connected to the Sunda-Andaman subduction system; the group is not simply a chain of coral atolls. The other options describe different processes, locations, scales or governance categories.

Q54. Which option is the safest spatial interpretation of Andaman-Nicobar origin?

A. Official 2026 PIB material describes an international container transshipment port, airport, township and power infrastructure as the proposed integrated project components. B. The Andaman and Nicobar Islands are emergent parts of a tectonic island arc connected to the Sunda-Andaman subduction system; the group is not simply a chain of coral atolls. C. The Ten Degree Channel separates the Andaman group from the Nicobar group, while the Nine Degree Channel separates Minicoy from the main Lakshadweep group. D. Great Nicobar is in UNESCO's World Network of Biosphere Reserves, added in 2013; biosphere-reserve recognition does not make the entire island an inviolate no-use zone.

Answer: B. Explanation: The Andaman and Nicobar Islands are emergent parts of a tectonic island arc connected to the Sunda-Andaman subduction system; the group is not simply a chain of coral atolls. The other options describe different processes, locations, scales or governance categories.

Q55. Which statement preserves the process boundary for Andaman-Nicobar origin?

A. Official 2026 PIB material describes an international container transshipment port, airport, township and power infrastructure as the proposed integrated project components. B. A Rajya Sabha answer dated 12 February 2026 stated that Stage-II or final forest approval had not been granted and project activity depended on final approval; a factsheet is not proof of completed mitigation. C. The Andaman and Nicobar Islands are emergent parts of a tectonic island arc connected to the Sunda-Andaman subduction system; the group is not simply a chain of coral atolls. D. Great Nicobar is in UNESCO's World Network of Biosphere Reserves, added in 2013; biosphere-reserve recognition does not make the entire island an inviolate no-use zone.

Answer: C. Explanation: The Andaman and Nicobar Islands are emergent parts of a tectonic island arc connected to the Sunda-Andaman subduction system; the group is not simply a chain of coral atolls. The other options describe different processes, locations, scales or governance categories.

Q56. Which option avoids the main UPSC trap concerning Andaman-Nicobar origin?

A. A Rajya Sabha answer dated 12 February 2026 stated that Stage-II or final forest approval had not been granted and project activity depended on final approval; a factsheet is not proof of completed mitigation. B. Island decisions must combine tectonic and tsunami exposure, reefs and nesting beaches, freshwater and sediment limits, indigenous rights, cumulative carrying capacity, alternatives and monitored safeguards. C. Official 2026 PIB material describes an international container transshipment port, airport, township and power infrastructure as the proposed integrated project components. D. The Andaman and Nicobar Islands are emergent parts of a tectonic island arc connected to the Sunda-Andaman subduction system; the group is not simply a chain of coral atolls.

Answer: D. Explanation: The Andaman and Nicobar Islands are emergent parts of a tectonic island arc connected to the Sunda-Andaman subduction system; the group is not simply a chain of coral atolls. The other options describe different processes, locations, scales or governance categories.

Q57. Which statement correctly explains Lakshadweep origin?

A. Lakshadweep consists of low coral islands, reefs and atolls on submarine ridges in the Arabian Sea; reef-derived land and freshwater lenses make the islands physically sensitive. B. The Ten Degree Channel separates the Andaman group from the Nicobar group, while the Nine Degree Channel separates Minicoy from the main Lakshadweep group. C. Great Nicobar is in UNESCO's World Network of Biosphere Reserves, added in 2013; biosphere-reserve recognition does not make the entire island an inviolate no-use zone. D. Official 2026 PIB material describes an international container transshipment port, airport, township and power infrastructure as the proposed integrated project components.

Answer: A. Explanation: Lakshadweep consists of low coral islands, reefs and atolls on submarine ridges in the Arabian Sea; reef-derived land and freshwater lenses make the islands physically sensitive. The other options describe different processes, locations, scales or governance categories.

Q58. Which option is the safest spatial interpretation of Lakshadweep origin?

A. A Rajya Sabha answer dated 12 February 2026 stated that Stage-II or final forest approval had not been granted and project activity depended on final approval; a factsheet is not proof of completed mitigation. B. Lakshadweep consists of low coral islands, reefs and atolls on submarine ridges in the Arabian Sea; reef-derived land and freshwater lenses make the islands physically sensitive. C. Great Nicobar is in UNESCO's World Network of Biosphere Reserves, added in 2013; biosphere-reserve recognition does not make the entire island an inviolate no-use zone. D. Official 2026 PIB material describes an international container transshipment port, airport, township and power infrastructure as the proposed integrated project components.

Answer: B. Explanation: Lakshadweep consists of low coral islands, reefs and atolls on submarine ridges in the Arabian Sea; reef-derived land and freshwater lenses make the islands physically sensitive. The other options describe different processes, locations, scales or governance categories.

Q59. Which statement preserves the process boundary for Lakshadweep origin?

A. Official 2026 PIB material describes an international container transshipment port, airport, township and power infrastructure as the proposed integrated project components. B. Island decisions must combine tectonic and tsunami exposure, reefs and nesting beaches, freshwater and sediment limits, indigenous rights, cumulative carrying capacity, alternatives and monitored safeguards. C. Lakshadweep consists of low coral islands, reefs and atolls on submarine ridges in the Arabian Sea; reef-derived land and freshwater lenses make the islands physically sensitive. D. A Rajya Sabha answer dated 12 February 2026 stated that Stage-II or final forest approval had not been granted and project activity depended on final approval; a factsheet is not proof of completed mitigation.

Answer: C. Explanation: Lakshadweep consists of low coral islands, reefs and atolls on submarine ridges in the Arabian Sea; reef-derived land and freshwater lenses make the islands physically sensitive. The other options describe different processes, locations, scales or governance categories.

Q60. Which option avoids the main UPSC trap concerning Lakshadweep origin?

A. Island decisions must combine tectonic and tsunami exposure, reefs and nesting beaches, freshwater and sediment limits, indigenous rights, cumulative carrying capacity, alternatives and monitored safeguards. B. A Rajya Sabha answer dated 12 February 2026 stated that Stage-II or final forest approval had not been granted and project activity depended on final approval; a factsheet is not proof of completed mitigation. C. Islands may be continental fragments or shelf islands, oceanic volcanic islands, coral islands or depositional islands; classification must state the origin criterion. D. Lakshadweep consists of low coral islands, reefs and atolls on submarine ridges in the Arabian Sea; reef-derived land and freshwater lenses make the islands physically sensitive.

Answer: D. Explanation: Lakshadweep consists of low coral islands, reefs and atolls on submarine ridges in the Arabian Sea; reef-derived land and freshwater lenses make the islands physically sensitive. The other options describe different processes, locations, scales or governance categories.

Q61. Which statement correctly explains India island-channel map?

A. The Ten Degree Channel separates the Andaman group from the Nicobar group, while the Nine Degree Channel separates Minicoy from the main Lakshadweep group. B. A Rajya Sabha answer dated 12 February 2026 stated that Stage-II or final forest approval had not been granted and project activity depended on final approval; a factsheet is not proof of completed mitigation. C. Great Nicobar is in UNESCO's World Network of Biosphere Reserves, added in 2013; biosphere-reserve recognition does not make the entire island an inviolate no-use zone. D. Official 2026 PIB material describes an international container transshipment port, airport, township and power infrastructure as the proposed integrated project components.

Answer: A. Explanation: The Ten Degree Channel separates the Andaman group from the Nicobar group, while the Nine Degree Channel separates Minicoy from the main Lakshadweep group. The other options describe different processes, locations, scales or governance categories.

Q62. Which option is the safest spatial interpretation of India island-channel map?

A. Official 2026 PIB material describes an international container transshipment port, airport, township and power infrastructure as the proposed integrated project components. B. The Ten Degree Channel separates the Andaman group from the Nicobar group, while the Nine Degree Channel separates Minicoy from the main Lakshadweep group. C. Island decisions must combine tectonic and tsunami exposure, reefs and nesting beaches, freshwater and sediment limits, indigenous rights, cumulative carrying capacity, alternatives and monitored safeguards. D. A Rajya Sabha answer dated 12 February 2026 stated that Stage-II or final forest approval had not been granted and project activity depended on final approval; a factsheet is not proof of completed mitigation.

Answer: B. Explanation: The Ten Degree Channel separates the Andaman group from the Nicobar group, while the Nine Degree Channel separates Minicoy from the main Lakshadweep group. The other options describe different processes, locations, scales or governance categories.

Q63. Which statement preserves the process boundary for India island-channel map?

A. Island decisions must combine tectonic and tsunami exposure, reefs and nesting beaches, freshwater and sediment limits, indigenous rights, cumulative carrying capacity, alternatives and monitored safeguards. B. Islands may be continental fragments or shelf islands, oceanic volcanic islands, coral islands or depositional islands; classification must state the origin criterion. C. The Ten Degree Channel separates the Andaman group from the Nicobar group, while the Nine Degree Channel separates Minicoy from the main Lakshadweep group. D. A Rajya Sabha answer dated 12 February 2026 stated that Stage-II or final forest approval had not been granted and project activity depended on final approval; a factsheet is not proof of completed mitigation.

Answer: C. Explanation: The Ten Degree Channel separates the Andaman group from the Nicobar group, while the Nine Degree Channel separates Minicoy from the main Lakshadweep group. The other options describe different processes, locations, scales or governance categories.

Q64. Which option avoids the main UPSC trap concerning India island-channel map?

A. Island decisions must combine tectonic and tsunami exposure, reefs and nesting beaches, freshwater and sediment limits, indigenous rights, cumulative carrying capacity, alternatives and monitored safeguards. B. Continental islands share continental crustal or shelf affinity and may be separated by drowning, faulting or erosion; proximity to a continent alone does not prove origin. C. Islands may be continental fragments or shelf islands, oceanic volcanic islands, coral islands or depositional islands; classification must state the origin criterion. D. The Ten Degree Channel separates the Andaman group from the Nicobar group, while the Nine Degree Channel separates Minicoy from the main Lakshadweep group.

Answer: D. Explanation: The Ten Degree Channel separates the Andaman group from the Nicobar group, while the Nine Degree Channel separates Minicoy from the main Lakshadweep group. The other options describe different processes, locations, scales or governance categories.

Q65. Which statement correctly explains Great Nicobar biosphere status?

A. Great Nicobar is in UNESCO's World Network of Biosphere Reserves, added in 2013; biosphere-reserve recognition does not make the entire island an inviolate no-use zone. B. A Rajya Sabha answer dated 12 February 2026 stated that Stage-II or final forest approval had not been granted and project activity depended on final approval; a factsheet is not proof of completed mitigation. C. Official 2026 PIB material describes an international container transshipment port, airport, township and power infrastructure as the proposed integrated project components. D. Island decisions must combine tectonic and tsunami exposure, reefs and nesting beaches, freshwater and sediment limits, indigenous rights, cumulative carrying capacity, alternatives and monitored safeguards.

Answer: A. Explanation: Great Nicobar is in UNESCO's World Network of Biosphere Reserves, added in 2013; biosphere-reserve recognition does not make the entire island an inviolate no-use zone. The other options describe different processes, locations, scales or governance categories.

Q66. Which option is the safest spatial interpretation of Great Nicobar biosphere status?

A. A Rajya Sabha answer dated 12 February 2026 stated that Stage-II or final forest approval had not been granted and project activity depended on final approval; a factsheet is not proof of completed mitigation. B. Great Nicobar is in UNESCO's World Network of Biosphere Reserves, added in 2013; biosphere-reserve recognition does not make the entire island an inviolate no-use zone. C. Islands may be continental fragments or shelf islands, oceanic volcanic islands, coral islands or depositional islands; classification must state the origin criterion. D. Island decisions must combine tectonic and tsunami exposure, reefs and nesting beaches, freshwater and sediment limits, indigenous rights, cumulative carrying capacity, alternatives and monitored safeguards.

Answer: B. Explanation: Great Nicobar is in UNESCO's World Network of Biosphere Reserves, added in 2013; biosphere-reserve recognition does not make the entire island an inviolate no-use zone. The other options describe different processes, locations, scales or governance categories.

Q67. Which statement preserves the process boundary for Great Nicobar biosphere status?

A. Island decisions must combine tectonic and tsunami exposure, reefs and nesting beaches, freshwater and sediment limits, indigenous rights, cumulative carrying capacity, alternatives and monitored safeguards. B. Islands may be continental fragments or shelf islands, oceanic volcanic islands, coral islands or depositional islands; classification must state the origin criterion. C. Great Nicobar is in UNESCO's World Network of Biosphere Reserves, added in 2013; biosphere-reserve recognition does not make the entire island an inviolate no-use zone. D. Continental islands share continental crustal or shelf affinity and may be separated by drowning, faulting or erosion; proximity to a continent alone does not prove origin.

Answer: C. Explanation: Great Nicobar is in UNESCO's World Network of Biosphere Reserves, added in 2013; biosphere-reserve recognition does not make the entire island an inviolate no-use zone. The other options describe different processes, locations, scales or governance categories.

Q68. Which option avoids the main UPSC trap concerning Great Nicobar biosphere status?

A. Continental islands share continental crustal or shelf affinity and may be separated by drowning, faulting or erosion; proximity to a continent alone does not prove origin. B. Oceanic volcanic islands grow where magma builds edifices above sea level at plate boundaries or hotspots; volcanic origin does not imply current eruptive activity. C. Islands may be continental fragments or shelf islands, oceanic volcanic islands, coral islands or depositional islands; classification must state the origin criterion. D. Great Nicobar is in UNESCO's World Network of Biosphere Reserves, added in 2013; biosphere-reserve recognition does not make the entire island an inviolate no-use zone.

Answer: D. Explanation: Great Nicobar is in UNESCO's World Network of Biosphere Reserves, added in 2013; biosphere-reserve recognition does not make the entire island an inviolate no-use zone. The other options describe different processes, locations, scales or governance categories.

Q69. Which statement correctly explains Great Nicobar project components?

A. Official 2026 PIB material describes an international container transshipment port, airport, township and power infrastructure as the proposed integrated project components. B. A Rajya Sabha answer dated 12 February 2026 stated that Stage-II or final forest approval had not been granted and project activity depended on final approval; a factsheet is not proof of completed mitigation. C. Islands may be continental fragments or shelf islands, oceanic volcanic islands, coral islands or depositional islands; classification must state the origin criterion. D. Island decisions must combine tectonic and tsunami exposure, reefs and nesting beaches, freshwater and sediment limits, indigenous rights, cumulative carrying capacity, alternatives and monitored safeguards.

Answer: A. Explanation: Official 2026 PIB material describes an international container transshipment port, airport, township and power infrastructure as the proposed integrated project components. The other options describe different processes, locations, scales or governance categories.

Q70. Which option is the safest spatial interpretation of Great Nicobar project components?

A. Island decisions must combine tectonic and tsunami exposure, reefs and nesting beaches, freshwater and sediment limits, indigenous rights, cumulative carrying capacity, alternatives and monitored safeguards. B. Official 2026 PIB material describes an international container transshipment port, airport, township and power infrastructure as the proposed integrated project components. C. Islands may be continental fragments or shelf islands, oceanic volcanic islands, coral islands or depositional islands; classification must state the origin criterion. D. Continental islands share continental crustal or shelf affinity and may be separated by drowning, faulting or erosion; proximity to a continent alone does not prove origin.

Answer: B. Explanation: Official 2026 PIB material describes an international container transshipment port, airport, township and power infrastructure as the proposed integrated project components. The other options describe different processes, locations, scales or governance categories.

Q71. Which statement preserves the process boundary for Great Nicobar project components?

A. Continental islands share continental crustal or shelf affinity and may be separated by drowning, faulting or erosion; proximity to a continent alone does not prove origin. B. Islands may be continental fragments or shelf islands, oceanic volcanic islands, coral islands or depositional islands; classification must state the origin criterion. C. Official 2026 PIB material describes an international container transshipment port, airport, township and power infrastructure as the proposed integrated project components. D. Oceanic volcanic islands grow where magma builds edifices above sea level at plate boundaries or hotspots; volcanic origin does not imply current eruptive activity.

Answer: C. Explanation: Official 2026 PIB material describes an international container transshipment port, airport, township and power infrastructure as the proposed integrated project components. The other options describe different processes, locations, scales or governance categories.

Q72. Which option avoids the main UPSC trap concerning Great Nicobar project components?

A. Coral islands are low accumulations of reef-derived carbonate sediment built and reworked by organisms, waves and currents; the living reef and the island landform are related but distinct. B. Continental islands share continental crustal or shelf affinity and may be separated by drowning, faulting or erosion; proximity to a continent alone does not prove origin. C. Oceanic volcanic islands grow where magma builds edifices above sea level at plate boundaries or hotspots; volcanic origin does not imply current eruptive activity. D. Official 2026 PIB material describes an international container transshipment port, airport, township and power infrastructure as the proposed integrated project components.

Answer: D. Explanation: Official 2026 PIB material describes an international container transshipment port, airport, township and power infrastructure as the proposed integrated project components. The other options describe different processes, locations, scales or governance categories.

Q73. Which statement correctly explains Great Nicobar status boundary?

A. A Rajya Sabha answer dated 12 February 2026 stated that Stage-II or final forest approval had not been granted and project activity depended on final approval; a factsheet is not proof of completed mitigation. B. Islands may be continental fragments or shelf islands, oceanic volcanic islands, coral islands or depositional islands; classification must state the origin criterion. C. Continental islands share continental crustal or shelf affinity and may be separated by drowning, faulting or erosion; proximity to a continent alone does not prove origin. D. Island decisions must combine tectonic and tsunami exposure, reefs and nesting beaches, freshwater and sediment limits, indigenous rights, cumulative carrying capacity, alternatives and monitored safeguards.

Answer: A. Explanation: A Rajya Sabha answer dated 12 February 2026 stated that Stage-II or final forest approval had not been granted and project activity depended on final approval; a factsheet is not proof of completed mitigation. The other options describe different processes, locations, scales or governance categories.

Q74. Which option is the safest spatial interpretation of Great Nicobar status boundary?

A. Islands may be continental fragments or shelf islands, oceanic volcanic islands, coral islands or depositional islands; classification must state the origin criterion. B. A Rajya Sabha answer dated 12 February 2026 stated that Stage-II or final forest approval had not been granted and project activity depended on final approval; a factsheet is not proof of completed mitigation. C. Oceanic volcanic islands grow where magma builds edifices above sea level at plate boundaries or hotspots; volcanic origin does not imply current eruptive activity. D. Continental islands share continental crustal or shelf affinity and may be separated by drowning, faulting or erosion; proximity to a continent alone does not prove origin.

Answer: B. Explanation: A Rajya Sabha answer dated 12 February 2026 stated that Stage-II or final forest approval had not been granted and project activity depended on final approval; a factsheet is not proof of completed mitigation. The other options describe different processes, locations, scales or governance categories.

Q75. Which statement preserves the process boundary for Great Nicobar status boundary?

A. Coral islands are low accumulations of reef-derived carbonate sediment built and reworked by organisms, waves and currents; the living reef and the island landform are related but distinct. B. Continental islands share continental crustal or shelf affinity and may be separated by drowning, faulting or erosion; proximity to a continent alone does not prove origin. C. A Rajya Sabha answer dated 12 February 2026 stated that Stage-II or final forest approval had not been granted and project activity depended on final approval; a factsheet is not proof of completed mitigation. D. Oceanic volcanic islands grow where magma builds edifices above sea level at plate boundaries or hotspots; volcanic origin does not imply current eruptive activity.

Answer: C. Explanation: A Rajya Sabha answer dated 12 February 2026 stated that Stage-II or final forest approval had not been granted and project activity depended on final approval; a factsheet is not proof of completed mitigation. The other options describe different processes, locations, scales or governance categories.

Q76. Which option avoids the main UPSC trap concerning Great Nicobar status boundary?

A. Coral islands are low accumulations of reef-derived carbonate sediment built and reworked by organisms, waves and currents; the living reef and the island landform are related but distinct. B. Oceanic volcanic islands grow where magma builds edifices above sea level at plate boundaries or hotspots; volcanic origin does not imply current eruptive activity. C. Reef-building corals are animals that secrete calcium-carbonate skeletons and commonly host photosynthetic zooxanthellae, linking clear sunlit water to reef productivity. D. A Rajya Sabha answer dated 12 February 2026 stated that Stage-II or final forest approval had not been granted and project activity depended on final approval; a factsheet is not proof of completed mitigation.

Answer: D. Explanation: A Rajya Sabha answer dated 12 February 2026 stated that Stage-II or final forest approval had not been granted and project activity depended on final approval; a factsheet is not proof of completed mitigation. The other options describe different processes, locations, scales or governance categories.

Q77. Which statement correctly explains Island appraisal framework?

A. Island decisions must combine tectonic and tsunami exposure, reefs and nesting beaches, freshwater and sediment limits, indigenous rights, cumulative carrying capacity, alternatives and monitored safeguards. B. Oceanic volcanic islands grow where magma builds edifices above sea level at plate boundaries or hotspots; volcanic origin does not imply current eruptive activity. C. Islands may be continental fragments or shelf islands, oceanic volcanic islands, coral islands or depositional islands; classification must state the origin criterion. D. Continental islands share continental crustal or shelf affinity and may be separated by drowning, faulting or erosion; proximity to a continent alone does not prove origin.

Answer: A. Explanation: Island decisions must combine tectonic and tsunami exposure, reefs and nesting beaches, freshwater and sediment limits, indigenous rights, cumulative carrying capacity, alternatives and monitored safeguards. The other options describe different processes, locations, scales or governance categories.

Q78. Which option is the safest spatial interpretation of Island appraisal framework?

A. Oceanic volcanic islands grow where magma builds edifices above sea level at plate boundaries or hotspots; volcanic origin does not imply current eruptive activity. B. Island decisions must combine tectonic and tsunami exposure, reefs and nesting beaches, freshwater and sediment limits, indigenous rights, cumulative carrying capacity, alternatives and monitored safeguards. C. Coral islands are low accumulations of reef-derived carbonate sediment built and reworked by organisms, waves and currents; the living reef and the island landform are related but distinct. D. Continental islands share continental crustal or shelf affinity and may be separated by drowning, faulting or erosion; proximity to a continent alone does not prove origin.

Answer: B. Explanation: Island decisions must combine tectonic and tsunami exposure, reefs and nesting beaches, freshwater and sediment limits, indigenous rights, cumulative carrying capacity, alternatives and monitored safeguards. The other options describe different processes, locations, scales or governance categories.

Q79. Which statement preserves the process boundary for Island appraisal framework?

A. Coral islands are low accumulations of reef-derived carbonate sediment built and reworked by organisms, waves and currents; the living reef and the island landform are related but distinct. B. Reef-building corals are animals that secrete calcium-carbonate skeletons and commonly host photosynthetic zooxanthellae, linking clear sunlit water to reef productivity. C. Island decisions must combine tectonic and tsunami exposure, reefs and nesting beaches, freshwater and sediment limits, indigenous rights, cumulative carrying capacity, alternatives and monitored safeguards. D. Oceanic volcanic islands grow where magma builds edifices above sea level at plate boundaries or hotspots; volcanic origin does not imply current eruptive activity.

Answer: C. Explanation: Island decisions must combine tectonic and tsunami exposure, reefs and nesting beaches, freshwater and sediment limits, indigenous rights, cumulative carrying capacity, alternatives and monitored safeguards. The other options describe different processes, locations, scales or governance categories.

Q80. Which option avoids the main UPSC trap concerning Island appraisal framework?

A. Coral islands are low accumulations of reef-derived carbonate sediment built and reworked by organisms, waves and currents; the living reef and the island landform are related but distinct. B. Reef-building corals are animals that secrete calcium-carbonate skeletons and commonly host photosynthetic zooxanthellae, linking clear sunlit water to reef productivity. C. Most reef-building corals favour warm, shallow, clear, well-lit, normally saline and oxygenated water with limited sediment stress; local tolerances vary. D. Island decisions must combine tectonic and tsunami exposure, reefs and nesting beaches, freshwater and sediment limits, indigenous rights, cumulative carrying capacity, alternatives and monitored safeguards.

Answer: D. Explanation: Island decisions must combine tectonic and tsunami exposure, reefs and nesting beaches, freshwater and sediment limits, indigenous rights, cumulative carrying capacity, alternatives and monitored safeguards. The other options describe different processes, locations, scales or governance categories.

PYQS AND ANSWER PRACTICE

VERIFIED PYQ OWNERSHIP AUDIT

Verified direct routes are the 2018 Prelims coral-reef distribution and biodiversity demand and the 2019 GS-I global-warming impact on coral life. Barren Island, biorock and island-nation sea-level questions retain their separate routed owners and are not duplicated as direct PYQs.

OWNER PYQ LEDGER EXTRACTS

Historical PYQ Integration (2018-2023)

> **Status:** Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-MAINS-GS1-GS2-ESSAY-2018-2023.md`, `_PYQ-ROUTING-PRELIMS-2018-2023.md`. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2018, 2019
- **Paper(s):** GS-I, Prelims GS-I
- **Routed question demands:** 2

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	Prelims GS-I	87	Coral reef global distribution tropical waters and biodiversity	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2019	GS-I	4	Global warming impact on the coral life system	Assess with examples · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.

What this owner must now support

- Coral reef global distribution tropical waters and biodiversity
- Global warming impact on the coral life system

> The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

PYQ DEMAND CARD 1 - 2018 Prelims GS-I

Demand: Coral reef global distribution in tropical waters and its biodiversity significance.

Status: Verified routed objective demand; official key unavailable in the local ledger.

Model solution: Test each statement independently: reef-building corals are concentrated mainly in warm, shallow tropical-subtropical seas, but distribution is not universal; the Coral Triangle and Great Barrier Reef are major centres; reefs support high taxonomic diversity. Preserve the official option order and do not invent an answer letter.

Demand decoding: Treat “PYQ DEMAND CARD 1 - 2018 Prelims GS-I” as a process, location, scale, terminology and status problem. Verify every statement independently.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in “PYQ DEMAND CARD 1 - 2018 Prelims GS-I”.

Analytical body:

1. **Claim and named evidence:** Demand: Coral reef global distribution in tropical waters and its biodiversity significance. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control,

exception or source/date boundary.

2. **Claim and named evidence:** Status: Verified routed objective demand; official key unavailable in the local ledger. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in “PYQ DEMAND CARD 1 - 2018 Prelims GS-I”.

Executable exam-length answer / compression plan: Fix the phenomenon and spatial setting; trace the controlling mechanism; test direction, sequence, threshold and India/world anchor; eliminate the closest term, map or causation distractor.

Why this earns marks: It preserves exact process-to-pattern reasoning and prevents a familiar place-name or correlation from substituting for causation.

How to improve this answer: For “PYQ DEMAND CARD 1 - 2018 Prelims GS-I”, state why the nearest distractor fails on process, direction, scale, location, terminology, date or official status.

PYQ DEMAND CARD 2 - 2019 GS-I

Demand: Assess the impact of global warming on the coral life system with examples. (150 words)

Status: Verified routed direct Mains demand.

Model solution: Explain marine heat stress and bleaching, then trace mortality, habitat simplification, fisheries, tourism, carbonate sediment and coastal protection. Use Indian reef settings as examples without unsupported event counts. Qualify that bleaching can recover if stress is brief and local pressures are reduced.

Demand decoding: The directive **answer** requires a direct position on “PYQ DEMAND CARD 2 - 2019 GS-I”, all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in “PYQ DEMAND CARD 2 - 2019 GS-I”.

Analytical body:

1. **Claim and named evidence:** Demand: Assess the impact of global warming on the coral life system with examples. (150 words) **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in “PYQ DEMAND CARD 2 - 2019 GS-I”.

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For “PYQ DEMAND CARD 2 - 2019 GS-I”, replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL MAINS 1 - 10 MARKS

Question: Classify islands by origin and explain why present shape alone is inadequate. Answer in about 150 words.

Model thesis: Continental, volcanic, coral and depositional origins reflect different crustal and sedimentary histories that can converge on similar present shapes.

Claim -> named evidence -> analysis -> qualification:

- Islands may be continental fragments or shelf islands, oceanic volcanic islands, coral islands or depositional islands; classification must state the origin criterion.
- Continental islands share continental crustal or shelf affinity and may be separated by drowning, faulting or erosion; proximity to a continent alone does not prove origin.
- Oceanic volcanic islands grow where magma builds edifices above sea level at plate boundaries or hotspots; volcanic origin does not imply current eruptive activity.
- Coral islands are low accumulations of reef-derived carbonate sediment built and reworked by organisms, waves and currents; the living reef and the island landform are related but distinct.

Qualified conclusion: Continental, volcanic, coral and depositional origins reflect different crustal and sedimentary histories that can converge on similar present shapes.

Demand decoding: The directive **explain** requires a direct position on “Classify islands by origin and explain why present shape alone is inadequate. Answer in about…”, all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Continental, volcanic, coral and depositional origins reflect different crustal and sedimentary histories that can converge on similar present shapes.

Analytical body:

1. **Claim and named evidence:** Islands may be continental fragments or shelf islands, oceanic volcanic islands, coral islands or depositional islands; classification must state the origin criterion. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Continental islands share continental crustal or shelf affinity and may be separated by drowning, faulting or erosion; proximity to a continent alone does not prove origin. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** Oceanic volcanic islands grow where magma builds edifices above sea level at plate boundaries or hotspots; volcanic origin does not imply current eruptive activity. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** Coral islands are low accumulations of reef-derived carbonate sediment built and reworked by organisms, waves and currents; the living reef and the island landform are related but distinct. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: Continental, volcanic, coral and depositional origins reflect different crustal and sedimentary histories that can converge on similar present shapes.

Executable exam-length answer / compression plan: For a 10-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise three process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For “Classify islands by origin and explain why present shape alone is inadequate. Answer in about…”, replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL MAINS 2 - 10 MARKS

Question: Differentiate fringing reefs, barrier reefs and atolls using a map-view sketch. Answer in about 150 words.

Model thesis: Distance from land, lagoon width and the presence or absence of a central high island separate the three forms.

Claim -> named evidence -> analysis -> qualification:

- A fringing reef lies close to shore with no lagoon or only a narrow shallow lagoon, making it the closest reef type to the land.
- A barrier reef lies farther offshore and is separated from land by a broader, deeper lagoon; offshore position and lagoon width distinguish it from a fringing reef.
- An atoll is a roughly ring-shaped reef or reef-island system enclosing a lagoon, generally without a central high island at the surface.

Qualified conclusion: Distance from land, lagoon width and the presence or absence of a central high island separate the three forms.

Demand decoding: The directive **answer** requires a direct position on “Differentiate fringing reefs, barrier reefs and atolls using a map-view sketch. Answer in…”, all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Distance from land, lagoon width and the presence or absence of a central high island separate the three forms.

Analytical body:

1. **Claim and named evidence:** A fringing reef lies close to shore with no lagoon or only a narrow shallow lagoon, making it the closest reef type to the land. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** A barrier reef lies farther offshore and is separated from land by a broader, deeper lagoon; offshore position and lagoon width distinguish it from a fringing reef. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** An atoll is a roughly ring-shaped reef or reef-island system enclosing a lagoon, generally without a central high island at the surface. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: Distance from land, lagoon width and the presence or absence of a central high island separate the three forms.

Executable exam-length answer / compression plan: For a 10-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise three process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For “Differentiate fringing reefs, barrier reefs and atolls using a map-view sketch. Answer in…”, replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL MAINS 3 - 15 MARKS

Question: Explain Darwin's reef-subsidence sequence and assess its limits. Answer in about 250 words.

Model thesis: Subsidence with upward-outward coral growth explains a powerful sequence, but sea-level oscillation, antecedent topography, ecology and tectonics require qualification.

Claim -> named evidence -> analysis -> qualification:

- Darwin's model links fringing reef to barrier reef to atoll as a volcanic foundation slowly subsides while coral growth keeps pace near sea level; it is a process model, not a rule for every reef.
- Daly emphasised lowered glacial sea level, wave planation and post-glacial flooding as controls on reef platforms; it supplements rather than erases biological growth and tectonics.

Qualified conclusion: Subsidence with upward-outward coral growth explains a powerful sequence, but sea-level oscillation, antecedent topography, ecology and tectonics require qualification.

Demand decoding: The directive **explain** requires a direct position on “Explain Darwin's reef-subsidence sequence and assess its limits. Answer in about 250 words.”, all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Subsidence with upward-outward coral growth explains a powerful sequence, but sea-level oscillation, antecedent topography, ecology and tectonics require qualification.

Analytical body:

1. **Claim and named evidence:** Darwin's model links fringing reef to barrier reef to atoll as a volcanic foundation slowly subsides while coral growth keeps pace near sea level; it is a process model, not a rule for every reef. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Daly emphasised lowered glacial sea level, wave planation and post-glacial flooding as controls on reef platforms; it supplements rather than erases biological growth and tectonics. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: Subsidence with upward-outward coral growth explains a powerful sequence, but sea-level oscillation, antecedent topography, ecology and tectonics require qualification.

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For “Explain Darwin's reef-subsidence sequence and assess its limits. Answer in about 250 words.”, replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL MAINS 4 - 15 MARKS

Question: Compare the geological and ecological character of Andaman-Nicobar and Lakshadweep. Answer in about 250 words.

Model thesis: A tectonic island arc and low coral-atoll system create different relief, hazards and resource limits, though both contain vulnerable reefs and coasts.

Claim -> named evidence -> analysis -> qualification:

- The Andaman and Nicobar Islands are emergent parts of a tectonic island arc connected to the Sunda-Andaman subduction system; the group is not simply a chain of coral atolls.
- Lakshadweep consists of low coral islands, reefs and atolls on submarine ridges in the Arabian Sea; reef-derived land and freshwater lenses make the islands physically sensitive.
- The Ten Degree Channel separates the Andaman group from the Nicobar group, while the Nine Degree Channel separates Minicoy from the main Lakshadweep group.

Qualified conclusion: A tectonic island arc and low coral-atoll system create different relief, hazards and resource limits, though both contain vulnerable reefs and coasts.

Demand decoding: The directive **compare** requires a direct position on “Compare the geological and ecological character of Andaman-Nicobar and Lakshadweep. Answer in…”, all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: A tectonic island arc and low coral-atoll system create different relief, hazards and resource limits, though both contain vulnerable reefs and coasts.

Analytical body:

1. **Claim and named evidence:** The Andaman and Nicobar Islands are emergent parts of a tectonic island arc connected to the Sunda-Andaman subduction system; the group is not simply a chain of coral atolls. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Lakshadweep consists of low coral islands, reefs and atolls on submarine ridges in the Arabian Sea; reef-derived land and freshwater lenses make the islands physically sensitive. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** The Ten Degree Channel separates the Andaman group from the Nicobar group, while the Nine Degree Channel separates Minicoy from the main Lakshadweep group. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: A tectonic island arc and low coral-atoll system create different relief, hazards and resource limits, though both contain vulnerable reefs and coasts.

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For “Compare the geological and ecological character of Andaman-Nicobar and Lakshadweep. Answer in…”, replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL MAINS 5 - 20 MARKS

Question: Assess how marine heat stress can transform coral-reef ecology, island security and livelihoods. Answer in about 300 words.

Model thesis: Bleaching affects habitat, fisheries, tourism, carbonate sediment and wave attenuation, with outcomes shaped by duration, local stress and recovery capacity.

Claim -> named evidence -> analysis -> qualification:

- Reef-building corals are animals that secrete calcium-carbonate skeletons and commonly host photosynthetic zooxanthellae, linking clear sunlit water to reef productivity.
- Most reef-building corals favour warm, shallow, clear, well-lit, normally saline and oxygenated water with limited sediment stress; local tolerances vary.
- Coral bleaching is the loss of symbiotic algae or pigments under stress, often marine heat; bleaching raises mortality risk but does not mean every colony is already dead.
- Reefs support biodiversity, fisheries, tourism, carbonate sediment and wave-energy attenuation, but they cannot eliminate extreme surge, tsunami or poor land-use exposure.

Qualified conclusion: Bleaching affects habitat, fisheries, tourism, carbonate sediment and wave attenuation, with outcomes shaped by duration, local stress and recovery capacity.

Demand decoding: The directive **assess** requires a direct position on “Assess how marine heat stress can transform coral-reef ecology, island security and…”, all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Bleaching affects habitat, fisheries, tourism, carbonate sediment and wave attenuation, with outcomes shaped by duration, local stress and recovery capacity.

Analytical body:

1. **Claim and named evidence:** Reef-building corals are animals that secrete calcium-carbonate skeletons and commonly host photosynthetic zooxanthellae, linking clear sunlit water to reef productivity. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Most reef-building corals favour warm, shallow, clear, well-lit, normally saline and oxygenated water with limited sediment stress; local tolerances vary. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** Coral bleaching is the loss of symbiotic algae or pigments under stress, often marine heat; bleaching raises mortality risk but does not mean every colony is already dead. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** Reefs support biodiversity, fisheries, tourism, carbonate sediment and wave-energy attenuation, but they cannot eliminate extreme surge, tsunami or poor land-use exposure. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: Bleaching affects habitat, fisheries, tourism, carbonate sediment and wave attenuation, with outcomes shaped by duration, local stress and recovery capacity.

Executable exam-length answer / compression plan: For a 20-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise six to eight process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For "Assess how marine heat stress can transform coral-reef ecology, island security and…", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL MAINS 6 - 20 MARKS

Question: Construct a balanced appraisal framework for the Great Nicobar project. Answer in about 300 words.

Model thesis: Separate official components and approval status from projected benefits, then test cumulative ecology, hazards, rights, carrying capacity, alternatives, safeguards and monitoring.

Claim -> named evidence -> analysis -> qualification:

- Great Nicobar is in UNESCO's World Network of Biosphere Reserves, added in 2013; biosphere-reserve recognition does not make the entire island an inviolate no-use zone.
- Official 2026 PIB material describes an international container transshipment port, airport, township and power infrastructure as the proposed integrated project components.
- A Rajya Sabha answer dated 12 February 2026 stated that Stage-II or final forest approval had not been granted and project activity depended on final approval; a factsheet is not proof of completed mitigation.
- Island decisions must combine tectonic and tsunami exposure, reefs and nesting beaches, freshwater and sediment limits, indigenous rights, cumulative carrying capacity, alternatives and monitored safeguards.

Qualified conclusion: Separate official components and approval status from projected benefits, then test cumulative ecology, hazards, rights, carrying capacity, alternatives, safeguards and monitoring.

Demand decoding: The directive **answer** requires a direct position on "Construct a balanced appraisal framework for the Great Nicobar project. Answer in about 300…", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Separate official components and approval status from projected benefits, then test cumulative ecology, hazards, rights, carrying capacity, alternatives, safeguards and monitoring.

Analytical body:

1. **Claim and named evidence:** Great Nicobar is in UNESCO's World Network of Biosphere Reserves, added in 2013; biosphere-reserve recognition does not make the entire island an inviolate no-use zone. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Official 2026 PIB material describes an international container transshipment port, airport, township and power infrastructure as the proposed integrated project components. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** A Rajya Sabha answer dated 12 February 2026 stated that Stage-II or final forest approval had not been granted and project activity depended on final approval; a factsheet is not proof of completed mitigation. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** Island decisions must combine tectonic and tsunami exposure, reefs and nesting beaches, freshwater and sediment limits, indigenous rights, cumulative carrying capacity, alternatives and monitored safeguards. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and

connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: Separate official components and approval status from projected benefits, then test cumulative ecology, hazards, rights, carrying capacity, alternatives, safeguards and monitoring.

Executable exam-length answer / compression plan: For a 20-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise six to eight process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For "Construct a balanced appraisal framework for the Great Nicobar project. Answer in about 300…", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.